

	University of Central Punjab Faculty of Information Technology & Computer Sciences Data Structures, Spring 2026 Quiz 5 V1	
	Total Marks: 20	Time Allowed: 30 min
Section	Student Name	Student Registration
D5		

CLO #	CLO Statement	Taxonomy Level	PLO
3	Analyze algorithms to understand their structure and functionality.	C5	3

Question # 1: Insert:

100, 50, 150, 25, 75, 125, 175, 10, 35, 60, 30, 55, 65, 58

Tasks

1. Draw the self balancing tree after every insertion.
2. Identify the first node whose BF becomes invalid.
3. Count:
 - o Total LL rotations
 - o Total RR rotations
 - o Total LR rotations
 - o Total RL rotations
4. Give the final balanced tree.
5. Give the final preorder traversal.

Question # 2: Given:

43, 12, 87, 25, 66, 9, 91, 30, 55

Tasks

1. Build Max Heap.
2. Show every heapify call.
3. Show every swap.
4. Show array after every extraction.
5. Count:
 - o Total heapify calls
 - o Total swaps
 - o Total comparisons